LEARNER GUIDE

# Growth models

Percentages, Ratio, Growth and Bounds · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

A **linear** model adds the same amount each step: $y = a + dn$. Its differences are all the same and its graph is a straight line. An **exponential** model multiplies by the same number each step: $y = a \times m^n$. Its ratios (each value divided by the one before) are all the same and its graph curves. With a multiplier above 1, exponential growth always overtakes linear growth in the end.

## What you are learning to notice

- Test a table: are the differences the same (linear), or the ratios (exponential)?
- In $y = a + dn$, $a$ is the start and $d$ is added each step.
- In $y = a \times m^n$, $a$ is the start and each step multiplies by $m$.
- A multiplier between 0 and 1 gives exponential decay.
- Exponential growth may start behind, but it catches up and overtakes.

## Moving through it, one step at a time

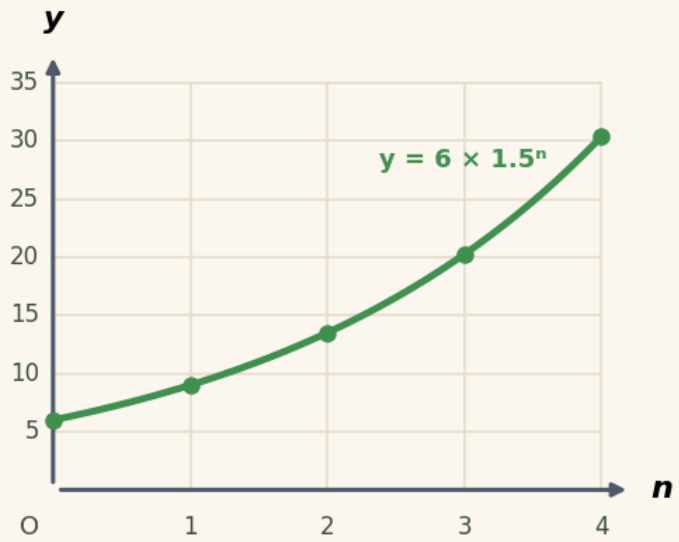
Write the values in a row. Under them write the differences, then the ratios. If the differences match, write $y$ = start + difference × $n$. If the ratios match, write $y$ = start × ratio$^n$. To find an overtake, make a table of both models and look for the first $n$ where the order changes.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

**Example 1 — Which model fits?**

A table goes 6, 9, 13.5, 20.25, … Is it linear or exponential? Find its rule and the next value.

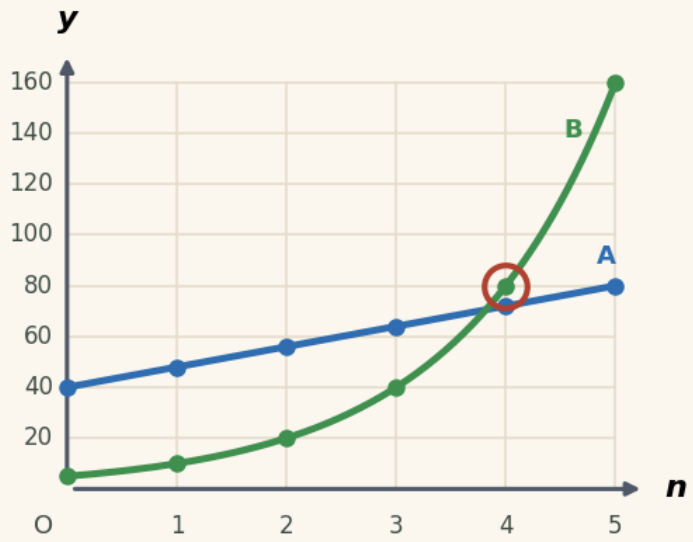



| | | |
|---|---|---|
| **1.** | The differences | 3, 4.5, 6.75: not the same |
| **2.** | The ratios | $\frac{9}{6} = 1.5$, $\frac{13.5}{9} = 1.5$, $\frac{20.25}{13.5} = 1.5$ |
| **3.** | The rule | $y = 6 \times 1.5^n$ |
| **4.** | The next value | $20.25 \times 1.5 = 30.375$ |

**Exponential growth, $y = 6 \times 1.5^n$; the next value is 30.375.**

**Example 2 — When does it overtake?**

Model A is $y = 40 + 8n$. Model B is $y = 5 \times 2^n$. At which $n$ does B first overtake A?




| | |
|---|---|
| **1.** $n = 2$ | A: 40 + 16 = 56; B: 5 × 4 = 20 |
| **2.** $n = 3$ | A: 64; B: 40 |
| **3.** $n = 4$ | A: 72; B: 80 |

**B first overtakes A at $n = 4$, and pulls further ahead every step after that.**

**A gentle reminder**

You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (straight-line graphs, sequences and repeated change (Lesson 6)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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